

ERP Project Documentation

Project: Enterprise Resource Planning (ERP) Suite

Tech Stack: React, ASP.NET Core, PostgreSQL, Local Storage, Amazon S3, Google Cloud Storage

1. Objective

Build a centralized ERP platform to manage Employees, Inventory, Vendors, Finance, File Storage, and Reports.

2. Modules

- Authentication & RBAC
- Employee Management
- Inventory Management
- Vendor Management
- Finance Module
- Document Management

3. Architecture

React UI → API Gateway → ASP.NET Core Backend → PostgreSQL / Redis / Local Storage / S3 / GCS

4. Database Tables

users(id, name, email, password_hash, role_id)

roles(id, role_name)

employees(id, employee_code, name, email, salary, department_id)

files(id, entity_type, entity_id, file_name, provider, path, created_at)

5. File Storage Providers

LOCAL: Development and testing

S3: Scalable object storage

GCS: Multi-region cloud storage

6. APIs

POST /auth/login

GET /employees

POST /employees

POST /files/upload

7. Security

JWT authentication, RBAC, BCrypt password hashing, upload validation, HTTPS.

8. Deployment

Docker, CI/CD via GitHub Actions or Jenkins, Monitoring via Grafana and Prometheus.

9. Milestones

Week 1-2: Setup + Auth + DB

Week 3-4: Employee module

Week 5-6: Inventory + Vendors

Week 7-8: File storage integration

Week 9-10: Reports + Deployment